

19) Decision Trees

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$$14) \quad G(Q) = 1 - \sum_c p_c^2$$

$$a) \quad G(\text{parent}) = 1 - \left(\frac{8}{14}\right)^2 - \left(\frac{6}{14}\right)^2$$

$$= 1 - \frac{64}{196} - \frac{36}{196} = 1 - \frac{100}{196} = \frac{96}{196}$$

$$\boxed{G(\text{parent}) = 0.490}$$

$$b) \quad G(\text{left}) = \text{~~scribbled out~~}$$

$$1 - \left(\frac{5}{6}\right)^2 - \left(\frac{1}{6}\right)^2 = 1 - \frac{25}{36} - \frac{1}{36} = \frac{10}{36}$$

$$\boxed{G(\text{left}) = 0.278}$$

$$G(\text{right}) = 1 - \left(\frac{3}{8}\right)^2 - \left(\frac{5}{8}\right)^2 = 1 - \frac{9}{64} - \frac{25}{64} =$$

$$\frac{30}{64} = \boxed{0.469 \quad G(\text{right})}$$

$$c) \quad G_{\text{split}} = \frac{n_L}{n} G(\text{left}) + \frac{n_R}{n} G(\text{right})$$

$$G_{\text{split}} = \frac{6}{14} \cdot 0.278 + \frac{8}{14} \cdot 0.469 =$$

$$0.119 + 0.268 = \boxed{0.387 \quad G_{\text{split}}}$$

$$G_{\text{gain}} = G(\text{parent}) - G_{\text{split}}$$

$$G_{\text{gain}} = 0.490 - 0.387 = \boxed{0.103 \quad G_{\text{gain}}}$$

1) Since $0.103 > 0.08$
the tree asks "does the
ticket contain 'urgent'?"

15) a) $\bar{y} = \frac{5+7+15+17+19+21}{6} = \frac{84}{6} = 14$
 $\bar{y} = 14$

$$SSE = \sum (y_i - \bar{y}_Q)^2$$

$$MSE(Q) = \frac{1}{n} \sum_{i \in Q} (y_i - \bar{y}_Q)^2 = \frac{SSE(Q)}{n}$$

$$\begin{aligned} SSE &= (5 - 14)^2 + (7 - 14)^2 + (15 - 14)^2 + (17 - 14)^2 + (19 - 14)^2 + (21 - 14)^2 \\ &= 81 + 49 + 1 + 9 + 25 + 49 = 214 \end{aligned}$$

214 $SSE(Q)$

$$MSE = \frac{214}{6} = \underline{\underline{35.67}}$$

b) $\bar{y}_{Left} = \frac{5+7}{2} = 6$ $y = 5, 7$

$$SSE_{left} = (5 - 6)^2 + (7 - 6)^2 = 2$$

$$MSE_{left} = \frac{2}{2} = 1$$

MSE_{left}

$$y = 15, 17, 19, 21$$

$$\bar{y} = \frac{15 + 17 + 19 + 21}{4} = \frac{72}{4} = 18$$

$$SSE_{\text{Right}} = (15 - 18)^2 + (17 - 18)^2 + (19 - 18)^2 + (21 - 18)^2$$

$$= 9 + 1 + 1 + 9 = 20 \quad \underline{\underline{SSE_{\text{Right}}}}$$

$$MSE_{\text{Right}} = \frac{20}{4} = 5$$

$$MSE_{\text{split}} = \frac{n_L}{n} MSE(L) + \frac{n_R}{n} MSE(R)$$

$$\text{Gain} = MSE_{\text{parent}} - MSE_{\text{split}}$$

$$MSE_{\text{split}} = \frac{2}{6} \cdot 1 + \frac{4}{6} \cdot 5 = 0.33 + 3.33 = 3.67$$

$$\text{Gain}_A = 5.67 - 3.67 = \underline{\underline{2}}$$

c) $N_0 = y(5, 15)$

$$\bar{y}_{N_0} = \frac{5 + 15}{2} = 10$$

$$SSE = (5 - 10)^2 + (15 - 10)^2 = 25 + 25 = \underline{\underline{50}}$$

$$MSE = \frac{50}{2} = 25$$

$$yes = y(7, 17, 19, 21)$$

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$$\bar{y}_{yes} = \frac{7+17+19+21}{4} = \frac{64}{4} = 16$$

$$SS_E = (\overset{-9}{7-16})^2 + (\overset{1}{17-16})^2 + (\overset{3}{19-16})^2 + (\overset{5}{21-16})^2$$

$$= 81 + 1 + 9 + 25 = 116$$

$$MSE = \frac{116}{4} = 29$$

$$MSE_{split} = \frac{2}{6} \cdot 25 + \frac{4}{6} \cdot 29 = 8.33 + 19.33 = 27.67$$

$$gain = 35.67 - 27.67 = 8$$

d) Since $32 > 8$ the tree makes split A.

Each leaf predicts the mean of its rows

$X \leq 3$: 6 minutes

$X > 3$: 18 minutes